INCH-POUND
A-A-59377
01 February 1999
SUPERSEDING
MIL-C-53071
08 September 1987

COMMERCIAL ITEM DESCRIPTION

COUPLING ASSEMBLY, QUICK-DISCONNECT, SEXLESS TYPE

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE. This commercial item description covers the performance requirements for the quick-disconnect coupling assemblies, sexless type. The quick-disconnect, sexless couplings are used to provide a positive seal disconnect between components of an aircraft fuel servicing system. The sexless coupling assembly shall be either a valved (dry break) or valveless (wet break) straight or fitting configuration with a 2-or 3-inch sexless end connection on one end, and a 2-or 3-inch adapter end connection on the other end. The valved coupling assembly shall prevent leakage when the valve is in the closed position. The valveless coupling assembly shall not prevent leakage when the coupling half is disconnected from its attachment.

1.1 Classification. The coupling assemblies shall be of the following types and classes, as specified (see 6.1):

1.1.1 Type.

Type 1	Straight configuration
Type 2	Fitting configuration

1.1.2 Class.

Class A	Valved (dry break)
Class B	Valveless (wet break)

2. SALIENT CHARACTERISTICS

2.1 Materials. The materials selected for construction are the responsibility of the contractor. The materials shall be of sufficient durability to meet the operational and environmental requirements as specified herein. Materials used in the manufacture of the coupling assembly shall be resistant to all fluids as specified in ASTM D 910. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Whenever feasible, toxic chemicals, hazardous substances or ozone depleting chemicals (ODC) shall not be used.

Beneficial comments, recommendations, additions, deletions, and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive and Armaments Command, ATTN: AMSTA-TR-D/210, Warren, MI 48397-5000.

AMSC N/A FSC 4730

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2.1.1 Deterioration prevention and control. The coupling assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which the material is susceptible.

2.1.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion (see 6.4.3).

2.1.3 Cleaning, treatment, and painting. Unless otherwise specified (see 6.1), all surfaces of the coupling, normally painted internally and externally shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice. Unless otherwise specified, the top coat color shall be tan.

2.2 Physical characteristics.

2.2.1 Valved coupling assembly (Class A). The valve shall prevent leakage in the closed position when the end cap is removed or when connected. The valve device shall be large enough to allow ease of operation by personnel wearing arctic mittens. The valve handle shall not stick out beyond the valve body, which would allow it to be damaged by dragging the assembly through brush or rocks. The valve control shall be labeled with a double-ended arrow showing direction of operation at each end to indicate the functional result (i.e., open, close, etc.). One person shall be capable of connecting or disconnecting the sexless end connection by hand.

2.2.2 Valveless coupling assembly (Class B). The mating parts of the valveless coupling assembly shall provide a positive lock by engaging and sealing against any leakage during operation. One person shall be capable of connecting or disconnecting the sexless end connection by hand.

2.2.3 Drop. The coupling assembly shall be capable of withstanding rough handling by not suffering loss of structural integrity when dropped (see 4.3.2).

2.2.4 Strength. The coupling assembly shall be sufficiently rugged to withstand normal operating, handling and maintenance loads without distortion or failure (see 4.3.3).

2.2.5 Endurance. When tested in accordance with 4.3.4, the coupling assembly shall be sufficiently rugged to withstand 2,000 connections and disconnections (complete cycles) when wet and 500 complete cycles when dry without functional distortion or failure. After completion of this test, the coupling assembly shall be capable of successfully passing the pressure test of 2.3.4 (see 4.3.6).

2.2.6 Weight. The weight of a valved or valveless coupling assembly, a sexless end connection mated with an adapter end connection, shall be no more than four (4) pounds for a two-inch coupling or five (5) pounds for a three-inch coupling.

2.2.7 Adapter end connections. Camlock connections shall conform to the requirements of ASTM F 1122. Hose connections shall be reusable and sized to mate with 2.00- or 3.00-inch standard collapsible or noncollapsible hose conforming to MIL-PRF-370, type A (collapsible), B (noncollapsible) or D (noncollapsible, low temperature).

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2.2.8 Valve interlocks. A valve interlock shall be provided to prevent sexless connection or disconnection when valve is open.

2.3 Performance.

2.3.1 Operating pressure. The coupling assembly shall operate at pressures ranging from 12 inches Hg vacuum to 100 psig without binding, leaking, excessive force required to operate, or chattering (see 4.3.1).

2.3.2 Burst pressure. The coupling assemblies, when joined together, shall withstand an internal hydrostatic pressure of not less than 300 psi maintained for not less than 1 minute without damage or leaks (see 4.3.5).

2.3.3 Flow capacity. When the coupling assembly is joined together its pressure drop, with a 200-gallon per minute (gpm) flow rate of fuel, shall not be greater than listed in Table I.

TABLE I: Coupling assembly pressure drop

Coupling assembly Nominal size (in)	Pressure drop (psi)
2	3
3	.5

2.3.4 Hydrostatic pressure. The coupling assembly shall withstand an internal hydrostatic pressure of 300 pounds per square inch (psi) maintained for not less than 1 minute without damage or leaks. The coupling assembly shall also withstand an internal hydrostatic pressure of 10 psi maintained for not less than 1 minute without damage or leaks (see 4.3.6).

2.3.5 Leakage. When tested in accordance with 4.3.7, the coupling assembly shall withstand internal hydrostatic pressures ranging from 12 inches of mercury (Hg), vacuum +5 and 100 psi with no damage or leaks.

2.3.6 Spillage. When tested in accordance with 4.3.7.3, spillage from mated valve coupling assemblies, when disconnected with valves closed, shall not exceed 15 cubic centimeters of fluid for 2-inch couplings and 40 cubic centimeters for 3-inch couplings.

2.3.7 Swiveling torque. The torque required to swivel the coupling assembly shall not be greater than 240 inch-pounds (see 4.3.8).

2.3.8 Valve actuation force. The force required to actuate the valve lever shall not exceed 15 pounds for the 2-inch coupling and 25 pounds for the 3-inch coupling (see 4.3.9).

2.3.9 Electrical resistance. The coupling assembly electrical resistance shall not exceed 5 ohms at a potential of 1.5 volts.

2.4 Environmental.

2.4.1 Dust protection. When connected or disconnected and capped, the internal surfaces of the coupling assembly shall not be contaminated by blowing dust, sand, and other foreign particles. A removable pressure cap shall be incorporated and provide dust proof connections with the sexless end (see 4.3.10.1).

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2.4.2 Corrosion. When tested in accordance with 4.3.10.2, the coupling assembly shall withstand the effects of humidity and salt atmosphere.

2.4.3 Solar radiation. The coupling assembly shall not suffer loss of structural integrity when exposed to solar radiation in accordance with 4.3.10.3.

2.4.4 Spill prevention. A valve lever shall be incorporated to prevent inadvertent valve opening or closing.

2.4.5 Operating environment. The coupling assembly shall be capable of operating in an environment of -25 °F to 125 °F (see 4.3.10.4).

2.4.6 Storage environment. The nozzle shall not be damaged by storage at ambient temperatures between -65 °F to 155 °F (see 4.3.10.4).

2.5 Interface.

2.5.1 Fuel. The coupling assembly shall be compatible with fuel conforming to MIL-DTL-5624, grade JP-8.

2.5.2 Existing couplings. The coupling assembly shall interface with existing sexless quick-disconnect couplings conforming to MIL-C-53071, Coupling Assemblies, Quick-Disconnect, Sexless Type.

2.6 Safety. The coupling assembly shall incorporate safety features that will ensure personnel safety, minimize human error in the field, and prevent accidental spills. The coupling assembly shall ensure electrical continuity during fuel operations by incorporating a dedicated contact. The device shall be easily operable in any operating environment including quick shut-off capability while the system is pressurized and flowing. The valve operating device shall be conveniently located; however, it shall also prevent inadvertent valve openings or closings by locking in the fully open and fully closed position.

2.7 Static control. To control static electricity during fueling operation, an electrical continuity device shall be incorporated in the coupling assembly. This device shall provide continuity contact between mating sexless surfaces without the use of external conductors. Provisions shall also be made to achieve intimate contact between hose bonding conductor and sexless coupling assembly when assembling the hose to the sexless coupling assembly. Static control devices shall also provide control of static electricity with all existing sexless connectors.

2.8 Identification marking. Identification markings shall be permanent, legible, and shall include the manufacturer's identification code, the manufacturer's part number, the part or identification number (PIN), and the national stock number (NSN).

3. REGULATORY REQUIREMENTS

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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4. QUALITY ASSURANCE

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such compliance.

4.2 Inspection requirements. Bid samples, when required for CID items to ensure product quality, shall be as specified in the contract or order.

4.2.1 Test conditions. Unless otherwise specified herein (see 6.1), the following test conditions shall apply. The valved mated coupling assembly shall be installed in a flow test stand and functionally tested. Before starting fluid flow, the valve(s) in the coupling assemblies shall be opened and closed at least 10 times. Then an attempt shall be made to disconnect the mated coupling assembly sexless end connection. The coupling interlocks shall prevent disconnection.

4.2.2 Test fluids. Unless otherwise specified herein, the test fluid shall be water or solvent conforming to ASTM D 910.

4.3 Tests.

4.3.1 Operating pressure. With valves open, fluid shall be allowed to flow through with pressures ranging from 12 inches Hg vacuum to 100 psig without binding, leaking, excessive force required to operate, or chattering. Nonconformance to 2.41 shall constitute failure of this test.

4.3.2 Drop. The coupling assembly shall withstand rough handling by not suffering loss of structural integrity when dropped eight times from a height of six (6) feet onto a concrete floor. Nonconformance to 2.3.3 shall constitute failure of this test.

4.3.3 Strength. The coupling assembly shall be subjected to 200 pounds bending force applied at one end (in a direction 90 degrees to the longitudinal center line) while the other end is pressurized to 100 psi internal pressure. While maintaining the pressure, the load shall be applied and removed 100 times. There shall be no leakage during this test or damage that causes malfunctioning of the assembly as a result of this test. The test shall also be repeated at a vacuum of 12 inches of Hg with an internal pressure of 10 psi. At the conclusion of this test, the assembly shall be subjected to a leakage test as specified in 4.3.6. Nonconformance to 2.3.4 shall constitute failure of this test.

4.3.4 Endurance.

4.3.4.1 Wet. The valved mating coupling assembly shall be connected and disconnected in the normal manner 2000 times (complete cycle) while fuel pressure of 100 pounds per square inch is applied internally each time the connection is made. The valve operating force test in 4.3.6.5 and the swiveling torque test in 4.3.6.4 shall be conducted after each 200 cycle. At the completion of this test, the leakage test in 4.3.6 shall be conducted. One test cycle shall consist of:

a. Connect two coupling assemblies together
b. Open coupling assembly valve
c. Begin flow

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d. Rotate (swivel) alternately in each direction
e. After not less than 10 seconds of flow, close coupling assembly valves
f. Disconnect assembly

Nonconformance to 2.3.5 shall constitute failure of this test.

4.3.4.2 Dry. The dry endurance test shall be conducted in a manner similar to the wet endurance test, except that the duration of the test shall be 500 cycles under dry conditions. The valve operating force test in 4.3.6.5 shall be conducted upon completion of the dry endurance test. Following the swiveling torque test in 4.3.6.4, the leakage test in 4.3.6 shall be conducted. One cycle shall consist of:

a. Connect two coupling assemblies together
b. Rotate (swivel) alternately in each direction
c. Open and close valves
d. Disconnect assembly

Nonconformance to 2.3.5 shall constitute failure of this test.

4.3.5 Burst pressure test. The coupling assembly shall be subjected to a burst pressure of 300 pounds psi for not less than 1 minute. Any damage or leaks shall be cause for rejection.

4.3.6 Hydrostatic pressure. Two coupling halves shall be mated together into a coupling assembly and shall be subjected to two hydrostatic pressures. The first shall be a hydrostatic pressure test of 300 psi for not less than 1 minute. The second shall be a hydrostatic pressure test of 10 psi for not less than 1 minute. Any damage or leak after either test shall be cause for rejection. The valve mated coupling assembly shall be tested twice at each hydrostatic pressure, with valves open and valves closed and capped.

4.3.7 Leakage.

4.3.7.1 Connected. The connected coupling assembly shall be subjected to internal hydrostatic pressure of 12 inches Hg vacuum, +5 and +100 psi. The pressure shall be maintained at each increment for not less than 1 minute. There shall be no loss of vacuum during negative pressure or leakage during positive pressure. Nonconformance to 2.4.5 shall constitute failure of this test.

4.3.7.2 Disconnected. The valve coupling assembly with valve closed shall be subjected to the internal hydrostatic pressure test specified in 4.3.5. Nonconformance to 2.4.5 shall constitute failure of this test.

4.3.7.3 Spillage test. The valve mated coupling assembly shall be subjected to a fluid flow rate of not less than 50 gpm. First the downstream coupling valve shall be closed, then the upstream coupling assembly valve shall be closed, and the sexless mating ends disconnected. Combined fluid spillage at the sexless mating connections during disconnection greater than 15 cubic centimeters shall be cause for rejection for 2.00-inch couplings and 50 cubic centimeters for 3.00-inch couplings. Nonconformance to 2.4.6 shall constitute failure of this test.

4.3.8 Swiveling torque. The coupling assembly shall be rotated 360 degrees in each direction to measure the starting torque when the internal pressure is 0 and 100 psi. Torque required to

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start or to sustain rotation shall not be greater than 240 inch-pounds. Nonconformance to 2.4.7 shall constitute failure of this test.

4.3.9 Valve.

4.3.9.1 Valve actuation. With valves open, fluid shall be allowed to flow through with flows ranging from 0 to 200 gallons per minute in 50 gallon per minute increments. At each flow increment, each valve shall be closed and opened not less than 10 times without binding, excessive force being required, or chattering. Nonconformance to 2.4.8 shall constitute failure of this test.

4.3.9.2 Valve actuation force. The valve coupling assembly valve shall be opened and closed when the internal pressure is 0 and 100 psi. The force required to open or close the valve shall not be greater than 15 pounds for the 2-inch coupling and 25 pounds for the 3-inch coupling. Nonconformance to 2.4.8 shall constitute failure of this test.

4.3.10 Environmental.

4.3.10.1 Dust. The dust test shall be performed in accordance with MIL-STD-810, method 510.1. Immediately after this test, the assembly shall be inspected and there shall be no deterioration that might adversely affect subsequent operation. The assembly shall then be subjected to the valve actuation force test in 4.3.6.5, swiveling torque test in 4.3.6.4 and leakage test in 4.3.6. Nonconformance to 2.5.1 shall constitute failure of this test.

4.3.10.2 Accelerated corrosion. The coupling assembly, with all fuel ports open, shall be submerged in a 2.5 percent solution by weight of sodium chloride and water. The assembly shall then be drained for 30 seconds and immediately placed in a test chamber maintained at a temperature of 86 °F with a relative humidity of 100 percent for a period of 20 minutes. At the completion of the humidity exposure period, the unit shall be placed in an air oven and maintained at 130 °F for a period of 20 minutes. The assembly shall not be operated again until a total of 50 immersion, humidity, and drying cycles have been completed. After the 50th cycle, the assembly shall be washed with warm water to remove all the salt accumulations and then dried. There shall be no evidence of deterioration that might adversely affect subsequent operation. The valve assembly shall then be subjected to the valve actuation force test in 4.3.6.5, to the swiveling force test in 4.3.6.4, and to the leakage test in 4.3.6.

4.3.10.3 Solar radiation. The solar radiation test shall be performed in accordance with MIL-STD-810, method 505.2. Nonconformance to 2.5.3 shall constitute failure of this test.

4.3.10.4 Operating temperature.

4.3.10.4.1 Low temperature. The coupling shall be filled with fuel and placed in a low temperature chamber and maintained at -65 ±2 °F for 24 hours after the chamber temperature is stabilized. At the completion of the 24-hour soak, the coupling assembly shall be inspected for damage and degradation. The coupling assembly shall then be soaked at -25 ±2 °F for 24 hours after the chamber temperature has stabilized. After completion of the 24-hour soak, while at -25 °F, the assembly shall be swiveled in each direction 10 times. On valved assemblies the valve shall be opened and closed 10 times. Afterwards, the assembly shall be subjected to the leakage

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tests in 4.3.6 and the valve actuation force test in 4.3.6.5. Nonconformance to 2.5.5 and 2.5.6 shall constitute failure of this test.

4.3.10.4.2 High temperature. The coupling assembly shall be filled with and soaked in a container filled with fuel conforming to MIL-DTL-5624, grade JP-5, or JP-8, and then shall be placed in an oven maintained at 155 ±2 °F. To prevent boiling, the container should be closed with pressure maintained not greater than 15 pounds psi. After the oven temperature is stabilized, the assembly shall be soaked for 72 hours. At the completion of the 72-hour soak, the assembly shall be removed from the oven and air dried for 72 hours. The assembly shall then be subjected to the dry endurance test in 4.3.3.2 and the valve actuation force test of 4.3.6.5. Nonconformance to 2.5.5 and 2.5.6 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Ordering data. Acquisition documents will specify the following:

a. Title, number, and date of this publication.
b. Coupling assembly type and class required (see 1.1).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 6.2).
d. Time frame required for submission of bid samples, and any additional inspections required (see 4.2).
e. When cleaning, treatment, and painting required (see 2.1.3).
f. If test conditions are other than as specified (see 4.2.1).
g. Packaging required (see 5.1).
h. Top coat, if required, and color (see 2.2.3).

6.2 Source of documents.

6.2.1 Military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

6.2.2 ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428.

6.3 Bid samples. When a bid sample inspection is required, the bid sample(s)must be identical to the production model. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the bid sample results, and disposition of the bid sample.

6.4 Definitions.

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6.4.1 CID. An acronym for Commercial Item Description.

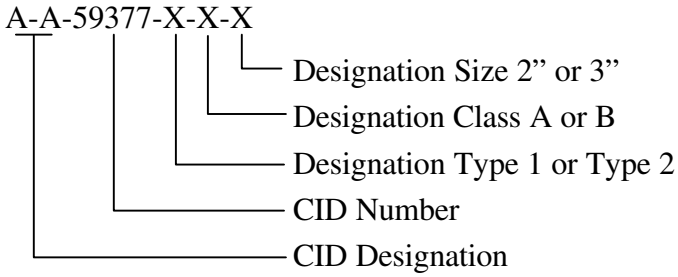
6.4.2 NSN. An acronym for National Stock Number.

6.4.3 Dissimilar metals. For its purpose, this document considers metals dissimilar when two specimens in contact with each other promote accelerated galvanic corrosion.

6.4.4 Recovered materials. For the purpose of this document, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials.

6.5 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for specification CID A-A-59377.



6.6 Data Requirements. The contracting officer shall include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor’s maintenance and operation manuals to be furnished with each clamp assembly. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12 L, Acquisition Management Systems and Data Requirements List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under a separate line item in the contract.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including special tools), and instructions on shipment of the sexless, quick-disconnect couplings.

6.8 Possible sources of products. Several manufacturers of products known to meet some or all of the requirements of this CID are listed below. Competition is not limited to these companies.

Aeroquip Corporation
Engineered Systems Group
Coupling Products Division
2500 W. Argyle
Jackson, Michigan 49202-1967

Carter Ground Fueling Co.
671 W. 17th Street
Costa Mesa, CA 92627

Custodian:

Preparing activity:

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Army – AT
Navy – SH
Air Force – 99
GSA – 7FXE

Review Activities:
DLA – CC

Army – AT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-59377	2. DOCUMENT DATE (YYMMDD) 990201

3. DOCUMENT TITLE
COUPLING ASSEMBLIES, QUICK-DISCONNECT, SEXLESS TYPE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TARDEC, AMSTA-TR-D/210 (Fuel & Water Handling Team)	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-7744 (2) AUTOVON DSN 786-7744
C. ADDRESS (Include Zip Code) US Army Tank-Automotive Research, Development & Engineering Center Attn: AMSTA-TR-D/210 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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